Lab 1 - Hashing

Name: Nathasa Bintang Kayesa

NIM: 2802438754

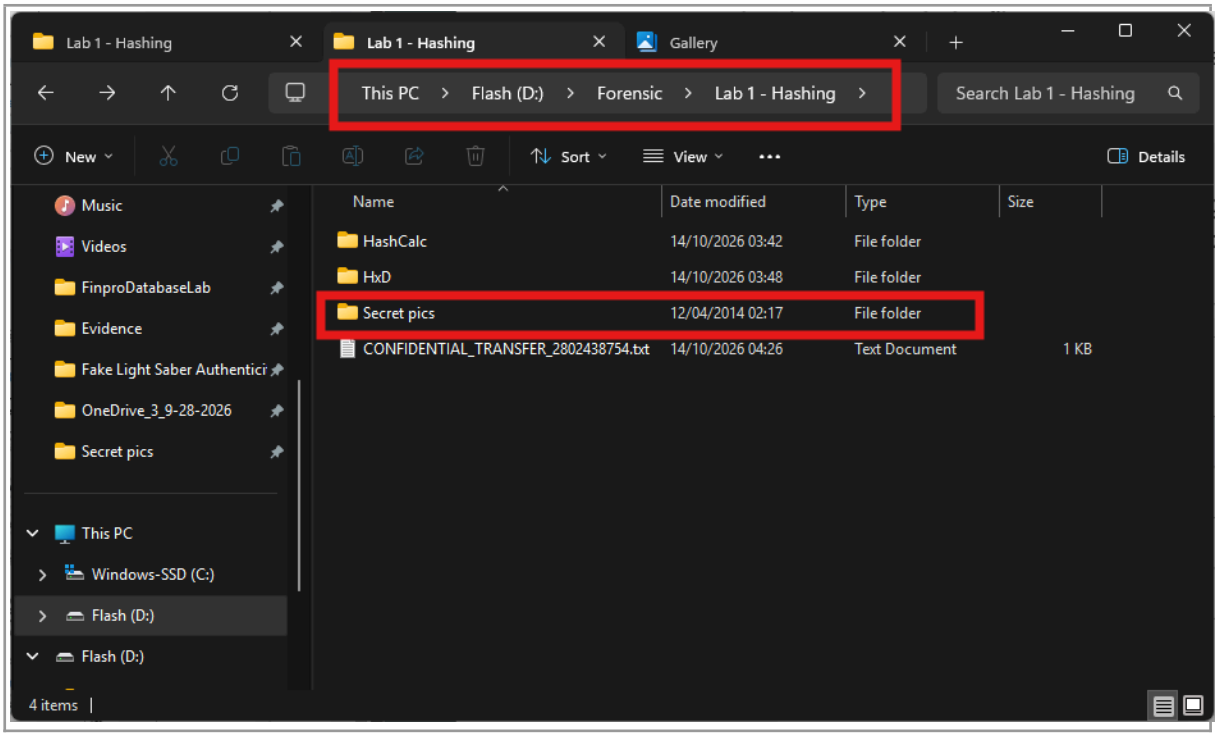
Date: 14/10/2026

Objectives

- Use HashCalc to determine the hash values of the files.
- Use HxD Hex Editor to change a single byte in a file.
- Use HashCalc to re-hash the files.
- Use HxD Hex Editor to examine the end of each file and determine the difference.

Steps 1-4: Add Evidence and Export Folder

Evidence image: Vader_Home_Computer.001 (FTK Imager). Path: C:\Documents and Settings\Owner\My Documents\Secret pics. The folder was exported to: D:\Forensic\Lab 1 - Hashing



Step 5: File Sizes

File	Size (bytes)
me & the guys1.jpg	251 KB (257.850 bytes)
me & the guys2.jpg	251 KB (257.850 bytes)

me & the guys3.jpg	251 KB (257.850 bytes)

Step 6: Description of Contents

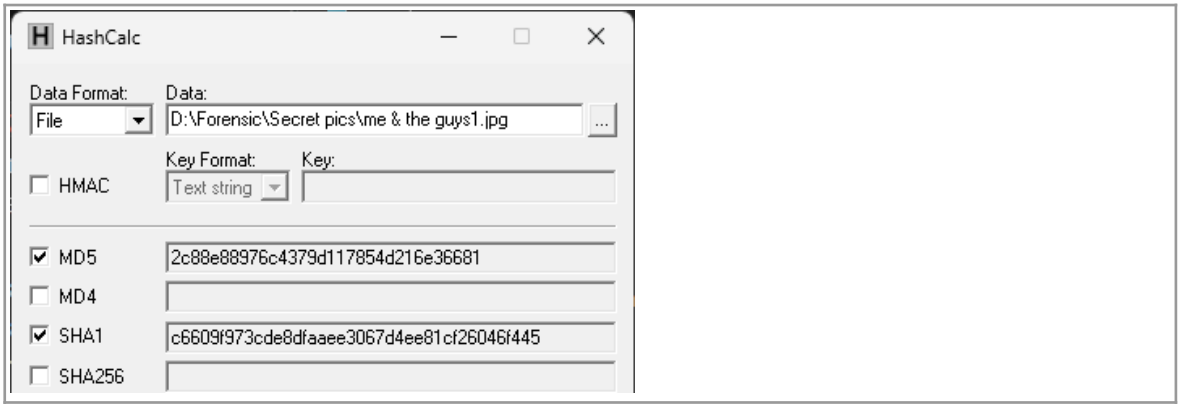
File	Description
me & the guys1.jpg	The image features about six people; one of them looks like a "Star Wars" character in black.
me & the guys2.jpg	Same as the previous image.
me & the guys3.jpg	Same as the previous image.

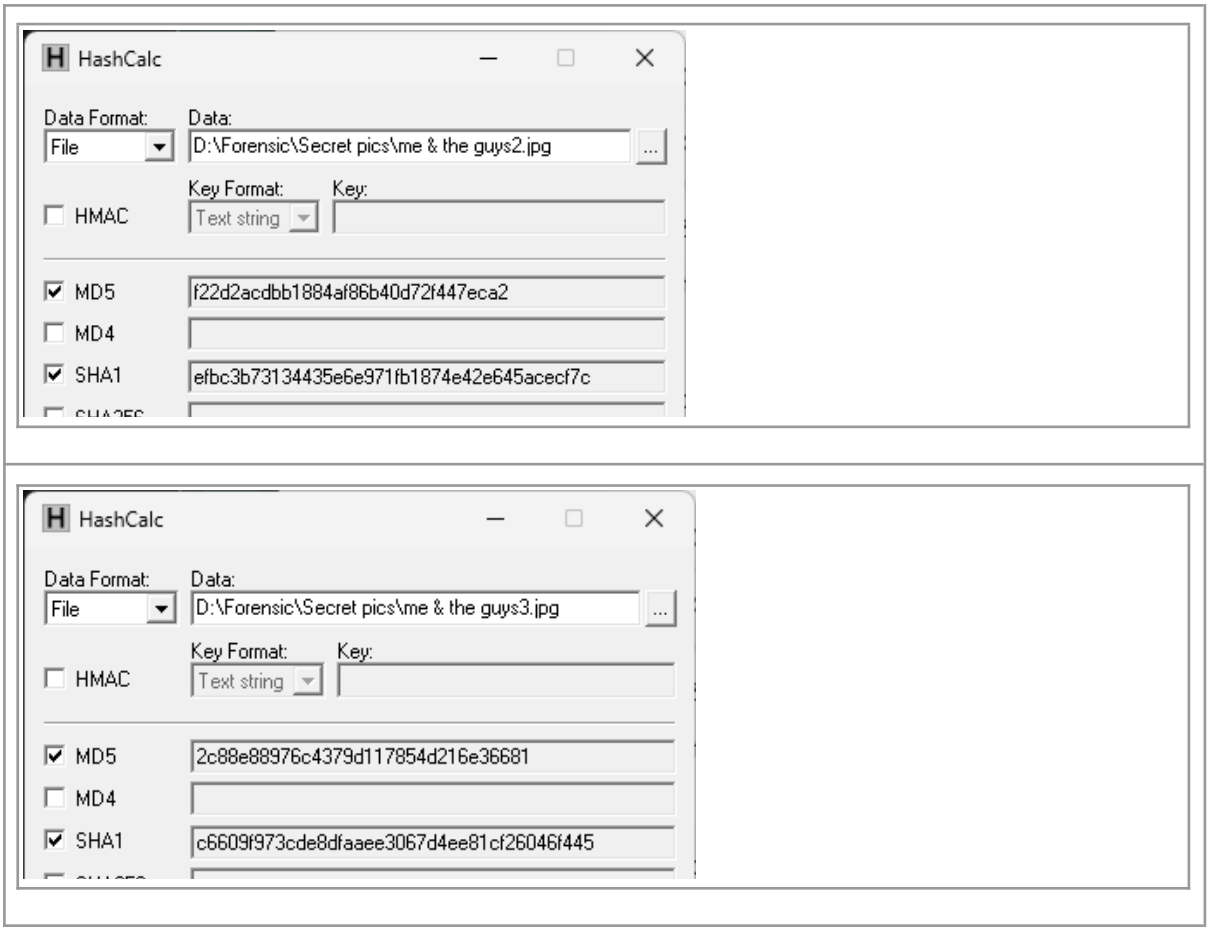
Step 7: Are the pictures all identical?

Answer: Yes, all of the images are identical. From the size, even the picture.

Steps 8-9: MD5 Hash Values (HashCalc)

File	MD5 Hash
me & the guys1.jpg	2c88e88976c4379d117854d216e36681
me & the guys2.jpg	f22d2acdbb1884af86b40d72f447eca2
me & the guys3.jpg	2c88e88976c4379d117854d216e36681





Duplicate files (identical hash): 2c88e88976c4379d117854d216e36681

Steps 10-13: Modify One Byte with HxD

File opened in HxD: Me & the guys1.jpg

Original last byte: D9 -> New last byte: 40

Saved as: Me & the guys1_modified.jpg

New File Result

Item	Value
Description	Image still opens and looks the same
Size	251 KB (257.850 bytes)
MD5 Hash	e7e5b6a5f7c5097a24640110212da9f5

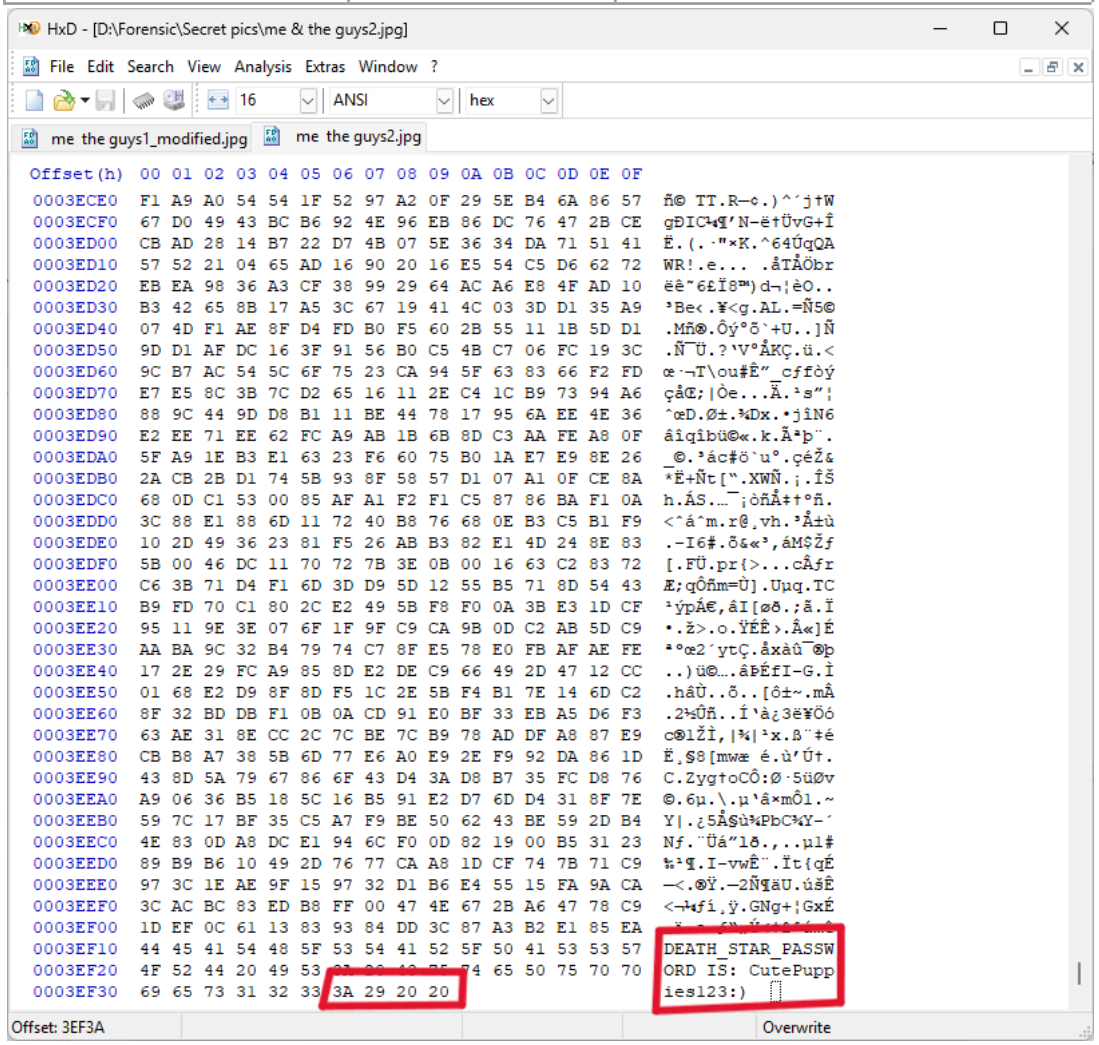
Question 14: Reliability of MD5 as a Digital Fingerprint

Based on the test, changing only one byte produced a completely different MD5 hash, while the file size and the visible image stayed the same. This shows that MD5 is very sensitive to any change in a file, so it is useful for verifying integrity

and detecting duplicate files. However, MD5 has known collision weaknesses, so for strong forensic evidence it should be combined with a stronger algorithm such as SHA-256.

Question 14 (second): Examining the End of Each File in HxD

File	Last bytes	Notes
me & the guys1.jpg	FF D9	
me & the guys2.jpg	20 20	Text found at end of file
me & the guys3.jpg	FF D9	



Finding: In me & the guys2.jpg, the end of the file contains ASCII text starting at offset 0003EF10: "DEATH_STAR_PASSWORD IS: CutePuppies123:)". This file does not contain the normal JPEG end marker (FF D9), while files 1 and 3 do. The text is not part of the image data and indicates hidden information.

Question 15: Can criminals hide information in a JPEG file?

Yes. Data can be placed at the end of a JPEG file (or overwrite its trailing bytes) and the picture still opens and looks normal, because viewers only render the image data. The hidden content is only visible when the file is inspected at the byte level, as done with HxD. It can be detected by comparing hash values, file sizes, and checking that the file structure ends with the correct FF D9 marker.

Conclusion

In this lab, MD5 hashes were calculated for three images exported from the Vader_Home_Computer.001 evidence image. Two of the files had identical hash values, showing they were duplicates, while the third differed. Changing only the last byte of a duplicate file produced a completely different MD5 hash even though the file size and the visible image stayed the same, which shows that hashing is very effective for verifying file integrity. Examining the end of each file in HxD revealed that me & the guys2.jpg did not end with the standard JPEG marker (FF D9) and contained the hidden text "DEATH_STAR_PASSWORD IS: CutePuppies123:)", while files 1 and 3 ended normally. This demonstrates that information can be hidden inside a JPEG file without changing how the picture looks, so forensic investigators need to check hash values, file sizes, and file structure at the byte level.